

UX/UI Case Study

TESTING AIDS

Problem

In the MAP Growth Student Testing application the test question aids had grown piecemeal, so each tool looked and behaved differently. Icons used mismatched styles, weights, and colors, and many didn't visually communicate what the tool did. The inconsistency pulled attention away from the test question, and students had to interpret the interface instead of focusing on the test question. Any solution had to work on Chromebooks, iPads, and desktops, meet WCAG 2.0 AA, and never distract from the question or compromise test integrity.

Role

Lead UX/UI Designer

Team

Project Manager
Engineering Team
Assessment Accessibility Lead

Tools

Illustrator
Sketch
Miro
Usability Sessions

RESEARCH

Conducted a visual audit of every aid, cataloging icon styles and documenting inconsistencies

Reviewed competitor assessment platforms and ran accessibility reviews

Ran icon recognition tests with subject matter experts and a small group of students to verify icon understanding

Key findings

Icons came from different visual languages and felt like separate products. Several icons were abstract and didn't depict the tool's function.

Final Example



Use the chart to answer the question.

Bikes at School	
March 12	    
March 13	
March 14	     
March 15	      

Each  represents 10 bicycles

On which date was there a total of 50 bicycles at school?

☐ A. March 12

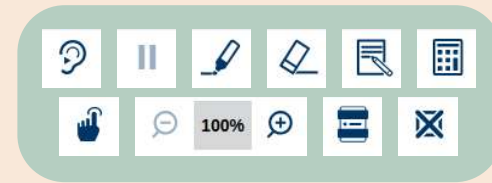
☐ B. March 13

☐ C. March 14

☐ D. March 15

 RESET Student Guest Practice Items: Calculator (Basic) Grade 4 Math K-12 Question # 1 

FRAMING



Define

Problem statement:

Students need tool icons that are instantly recognizable and visually quiet, because inconsistent, unclear icons distract from the question and add cognitive load during testing.

How We Might

Create one cohesive visual language across all aids?

Make each icon show what its tool does at a glance?

Keep the tools present but unobtrusive during an assessment?

Success Metrics

Icon recognition rate, time to identify a tool, consistency audit score, accessibility audit pass.



IDEATE & PROTOTYPE



Ideate

In a workshop with the PM, engineers, and assessment specialists, I explored three icon directions: realistic illustrations, simple outline glyphs, and filled pictograms. I chose a combined filled and outline style with consistent fills, strokes, and corner radii, where each icon depicts the tool's functional counterpart.

Prototype

I moved from low-fidelity sketches to a high-fidelity Sketch prototype and component library covering all button states and hover titles. I then built interactive prototypes of each tool so the icons could be evaluated in context, the way students would use them.

Key Decisions

A shared icon grid with fill and stroke specifications

Text labels as hover titles

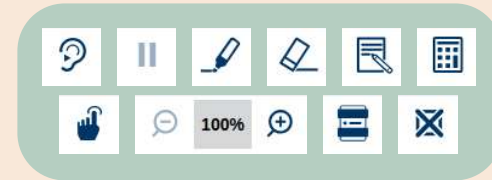
A consistent color palette so tools don't compete with the question

Buttons that follow the Design System guidelines

Accessibility Requirements:

Keyboard navigation, logical focus order, ARIA labels, and screen reader announcements

VALIDATE REFLECT



Round 1

Small
Student
Group

Icon recognition testing showed the eraser was still misread.

Result

Small change to erasure adding a line hinting at the action.

Line reader full functionality added

Round 2

Small
Student
Group

Feedback that Line Reader icon did not reflect added functionality

Result

Revised the height of the alignment bars in line reader icon to reflect the tool's final functionality

Pilot

District
Student
Group

Live Nebraska State Testing

Result

The redesigned aids were validated in a live pilot during Nebraska state testing, a high-stakes environment where interface problems surface quickly. No tool-related issues were reported by students, teachers, or proctors, and the design passed the WCAG 2.0 AA audit.